Université d'Ottawa · University of Ottawa
Faculté des sciences
Mathématiques et de statistique
Faculty of Science
Mathematics and Statistics

MAT 1302 B: Mathematical methods II

Instructor: Aziz Khanchi

Test I - Blue version
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Surname _____ First Name _____

Student # _____ DGD (1–4) _____

Instructions:

- No calculators or other electronic aids allowed.
- No notes, books or other papers allowed.
- Answer all questions in the space provided. You must justify your answers and explain your reasoning.
- There are 9 pages. In all there are 8 questions worth a total of 30 marks. There are 4 multiple choice questions, and 4 long answer questions. There will be no part marking for the multiple choice questions. For the long answer questions, you must justify your answers.

GOOD LUCK!

Answers to multiple choice Qs

1	2	3	4

585, av. King-Edward C.P. 450, Succ. A
Ottawa (Ontario) K1N 6N5 Canada
585 King Edward Ave., P.O. Box 450, Stn. A
Ottawa, Ontario K1N 6N5 Canada
(613) 562-5864 • Téléc./Fax (613) 562-5776
Courriel/Email: uomaths@science.uottawa.ca

Part 1: Multiple choice questions, 2 points each (no partial marks), chose the best possible answer.

Question 1. $\begin{bmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} =$

(a) $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, (b) $\begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$, (c) $\begin{bmatrix} 6 \\ 15 \end{bmatrix}$, (d) $\begin{bmatrix} 4 \\ 10 \\ 18 \end{bmatrix}$, (e) $\begin{bmatrix} 15 \\ 6 \end{bmatrix}$.

Solution: (e)

Question 2. Given (i) $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$, (ii) $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$, (iii) $\begin{bmatrix} 6 \\ 3 \end{bmatrix}$. Which of them are in the span$\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix} \right\}$?

(a) (i) and (ii), (b) (i) and (iii), (c) (ii) and (iii), (d) (i), (ii) and (iii), (e) (i) only.

Solution: (c)

Question 3. Given

$$A = \begin{bmatrix} 1 & 0 & 3 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \\ 2 & 0 & 1 & 1 \end{bmatrix}, D = \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{bmatrix}.$$

List all matrices which are in row reduced echelon form.

(a) A, B, and D, (b) A and B, (c) A and D, (d) D only, (e) B and D.

Solution: (d)

Question 4. Given that $\vec{x} = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$ is a solution of the non-homogeneous system $A\vec{x} = \vec{b}$, and that $t \begin{bmatrix} 2 \\ 1 \end{bmatrix}$ is the general solution of the corresponding homogeneous system $A\vec{x} = \vec{0}$, where $t \in \mathbb{R}$, which of the following is a solution of $A\vec{x} = \vec{b}$?

(a) $\begin{bmatrix} 3 \\ 2 \end{bmatrix}$; (b) $\begin{bmatrix} 4 \\ 0 \end{bmatrix}$; (c) $\begin{bmatrix} 3 \\ 1 \end{bmatrix}$; (d) $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$; (e) $\begin{bmatrix} 5 \\ 1 \end{bmatrix}$;

Solution: (c)

Part 2: Long Answer Questions (Details are necessary!)

Question 5. [**3**] Consider the following linear system

$$\begin{cases} x_1 - x_2 & = 2 \\ 2x_1 + hx_2 & = k. \end{cases}$$

(i) For what values of h and k does the system have no solution?

(ii) For what values of h and k does the system have only one solution?

(iii) For what values of h and k does the system have infinite solutions?

Solution: The augmented matrix of the system is

$$\left[\begin{array}{cc|c} 1 & -1 & 2 \\ 2 & h & k \end{array}\right] \xrightarrow{-2R_1+R_2\to R_2} \left[\begin{array}{cc|c} 1 & -1 & 2 \\ 0 & h+2 & k-4 \end{array}\right]$$

(i) $h = -2, k \neq 4$;
(ii) $h \neq -2, k \in \mathbb{R}$;
(iii) $h = -2, k = 4$;

Question 6. Consider the following system of linear equations

$$\begin{aligned} x_1 - 8x_3 + 3x_4 &= 2 \\ x_2 - 7x_3 - 3x_4 &= 1 \end{aligned}$$

(i) [**1**] Write the vector equation corresponding to the system?
(ii) [**1**] Write the matrix equation corresponding to the system?
(iii) [**2**] What is the augmented matrix of the system? Is it in reduced row echelon form?
(iii) [**4**] Solve the system and write the general solution in parametric vector form.

Solution: (i) The vector equation of the system:

$$x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} -8 \\ -7 \end{bmatrix} + x_4 \begin{bmatrix} 3 \\ -3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

(ii) The matrix equation of the system:

$$\begin{bmatrix} 1 & 0 & -8 & 3 \\ 0 & 1 & -7 & -3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

(iii)

$$\text{the augmented matrix} = \left[\begin{array}{cccc|c} 1 & 0 & -8 & 3 & 2 \\ 0 & 1 & -7 & -3 & 1 \end{array}\right].$$

It is in reduced row echelon form.

(iv) We have

$$\begin{aligned} x_1 &= 2 + 8x_3 - 3x_4 \\ x_2 &= 1 + 7x_3 + 3x_4 \\ x_3 &\quad free \\ x_4 &\quad free \end{aligned}$$

Let $x_3 = s$, $x_4 = t$, we have $x_1 = 2 + 8s - 3t$, $x_2 = 1 + 7s + 3t$. Thus

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 2 + 8s - 3t \\ 1 + 7s + 3t \\ s \\ t \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + s \begin{bmatrix} 8 \\ 7 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 3 \\ 0 \\ 1 \end{bmatrix}.$$

Question 7. **[5]** Given four vectors $\vec{u} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$, $\vec{v} = \begin{bmatrix} 0 \\ 8 \\ -2 \end{bmatrix}$, $\vec{w} = \begin{bmatrix} 6 \\ 5 \\ 5 \end{bmatrix}$, $\vec{y} = \begin{bmatrix} -2 \\ -15 \\ 7 \end{bmatrix}$. Write $\vec{y}$ as a linear combination of $\vec{u}, \vec{v}, \vec{w}$.

Solution:

$$\left[\begin{array}{ccc|c} \vec{u} & \vec{v} & \vec{w} & \vec{y} \end{array}\right] = \left[\begin{array}{ccc|c} 2 & 0 & 6 & -2 \\ -1 & 8 & 5 & -15 \\ 1 & -2 & 5 & 7 \end{array}\right] \xrightarrow{\frac{1}{2}R_1} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ -1 & 8 & 5 & -15 \\ 1 & -2 & 5 & 7 \end{array}\right]$$

$$\xrightarrow[-R_1+R_3\to R_3]{R_2+R_1\to R_2} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ 0 & 8 & 8 & -16 \\ 0 & -2 & 2 & 8 \end{array}\right] \xrightarrow{\frac{1}{8}R_2} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ 0 & 1 & 1 & -2 \\ 0 & -2 & 2 & 8 \end{array}\right] \xrightarrow{2R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 4 & 4 \end{array}\right]$$

$$\xrightarrow{\frac{1}{4}R_3} \left[\begin{array}{ccc|c} 1 & 0 & 3 & -1 \\ 0 & 1 & 1 & -2 \\ 0 & 0 & 1 & 1 \end{array}\right] \xrightarrow[-R_3+R_2\to R_2]{-3R_3+R_1\to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & -4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

The only solution of the system is $(-4, -3, 1)$. Therefore, $\vec{y}$ can be written as a linear combination of $\vec{u}, \vec{v}, \vec{w}$:

$$\vec{y} = -4\vec{u} - 3\vec{v} + \vec{w}.$$

Question 8. Consider the following system of linear equations

$$\begin{cases} x_1 + 4x_2 - 8x_3 &= 0 \\ 2x_1 + 5x_2 - 7x_3 &= 0 \\ -3x_1 - 7x_2 + 9x_3 &= 0 \end{cases}$$

(i) **[3]** Reduce the augmented matrix to the reduced row echelon form.
(ii) **[1]** Does the system have non-trivial solutions? Explain.
(iii) **[2]** Write the solution set in parametric vector form.

Solution:
(i)

$$\text{augmented matrix} = \left[\begin{array}{ccc|c} 1 & 4 & -8 & 0 \\ 2 & 5 & -7 & 0 \\ -3 & -7 & 9 & 0 \end{array}\right] \xrightarrow[R_3+3R_1\to R_3]{-2R_1+R_2\to R_2} \left[\begin{array}{ccc|c} 1 & 4 & -8 & 0 \\ 0 & -3 & 9 & 0 \\ 0 & 5 & -15 & 0 \end{array}\right]$$

$$\xrightarrow{-\frac{1}{3}R_2} \left[\begin{array}{ccc|c} 1 & 4 & -8 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 5 & -15 & 0 \end{array}\right] \xrightarrow{-5R_2+R_3\to R_3} \left[\begin{array}{ccc|c} 1 & 4 & -8 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-4R_2+R_1\to R_1} \left[\begin{array}{ccc|c} 1 & 0 & 4 & 0 \\ 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

(ii) Last column of the coefficient matrix is not pivot. Therefore, x_3 is a free variable and non-trivial solutions exist.

iii) From (i) we have

$$\begin{array}{rcl} x_1 + 4x_3 &=& 0 \\ x_2 - 3x_3 &=& 0 \\ x_3 && \text{free} \end{array}$$

i.e.,

$$\begin{array}{rcl} x_1 &=& -4t \\ x_2 &=& 3t \\ x_3 &=& t \text{ (free)} \end{array}$$

Then the general solution is:

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -4t \\ 3t \\ t \end{bmatrix} = t \begin{bmatrix} -4 \\ 3 \\ 1 \end{bmatrix}$$

$$\text{solution set} = \left\{ t \begin{bmatrix} -4 \\ 3 \\ 1 \end{bmatrix} \mid \quad t \in \mathbb{R} \right\}$$

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